



SHUBHAM KAKASAHEB GAIKWAD

Aspiring Robotics Design Engineer | Founder, Yantrigen Labs

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EDUCATION

B.E. Mechanical Engineering

Savitribai Phule Pune Univ. |
2026 – Present

Currently in 2nd Year.
Focus on bridging core mechanics with modern software automation.

Certifications

- Human Values and Professional Ethics (SWAYAM - 12 weeks)

TECH SKILLS

Robotics & Hardware

Arduino (UNO), MPU6050, LDR, GPS, Actuators (Servo/DC), PID Control, Sensors

Mechanical Design

Kinematics, CAD/CAM, Aerodynamic Balancing, Structural Stability Analysis

Software & AI

AI Chatbots (NLP basics), UI/UX, HTML/CSS, C/C++ Basics

PROFESSIONAL SUMMARY

2nd-year Mechanical Engineering undergraduate with a strong passion for automation, electromechanical systems, and artificial intelligence. Founder of **Yantrigen Labs**, a tech initiative dedicated to building advanced robotic prototypes and engineering software utilities. Proven ability to combine mechanical principles with software logic.

ENTREPRENEURIAL EXPERIENCE

Founder & Lead Engineer | Yantrigen Labs

Jalna, Maharashtra | 2026 – Present

• Brand Architecture:

Custom-coded (HTML/CSS) and deployed a fully responsive, dark-mode professional portfolio platform hosted on GitHub Pages.

• **Digital Identity:** Verified domain via Google Search Console and executed technical SEO to establish brand authority in the robotics space.

• Project Leadership:

Leading R&D of 6+ advanced engineering prototypes, managing mechanical CAD design, software logic, and UI/UX.

• Technical Documentation:

Authored comprehensive technical articles on kinematics and control systems to build a robust knowledge base.

KEY ENGINEERING PROJECTS

1. RoboArm Mk.I

Engineered a functional robotic arm for automated pick-and-place operations. Calculated precise joint angles using inverse kinematics to coordinate multi-axis motion via servo motors.

VISION

To clear the GATE exam, pursue an M.Tech in Robotics, and build scalable automated electromechanical systems.

2. Self-Balancing Robot

Developed an autonomous two-wheeled robot based on the inverted pendulum concept. Integrated an MPU6050 IMU sensor and a custom-tuned PID control algorithm.

3. Smart Solar Tracker

Designed a renewable energy automation system utilizing LDRs and an Arduino to continuously orient solar panels perpendicular to the sun for maximum efficiency.

4. Drone & UAV Systems

Prototyped UAVs focusing on the integration of flight dynamics, aerodynamics, and intelligent flight controllers for automated mapping.

5. AI Chatbot & MechScribe Utility

Applied Design Thinking to develop an AI-driven chatbot for small businesses. Designed UI/UX for MechScribe, a centralized app helping engineering students organize resources.